

Coordination Under Existential Unawareness

Information Ablation and Closure Thresholds in LLM Multi-Agent Systems

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Code Paper Data Available
<https://github.com/kehao95/quine>

Abstract

We placed LLM agents in a shared task that no agent can solve alone, then withheld every statement that collaboration, peers, or a communication channel exists. Agents can still discover one another from a changing world and build coordination through ordinary shared files. But the experiment separates two thresholds that are usually conflated: a static trace can make another agent imaginable; only a responsive environmental loop can keep a joint activity coordinated. Social knowledge is therefore not the prerequisite for sociality. Sustained coupling is.

Experimental Conditions: What the World Reveals

A task no agent can succeed alone:

collect 15 cells from a shared world. The global budget is 17; each agent can collect at most 10. No one can succeed alone. Agents can reset the budget and task.

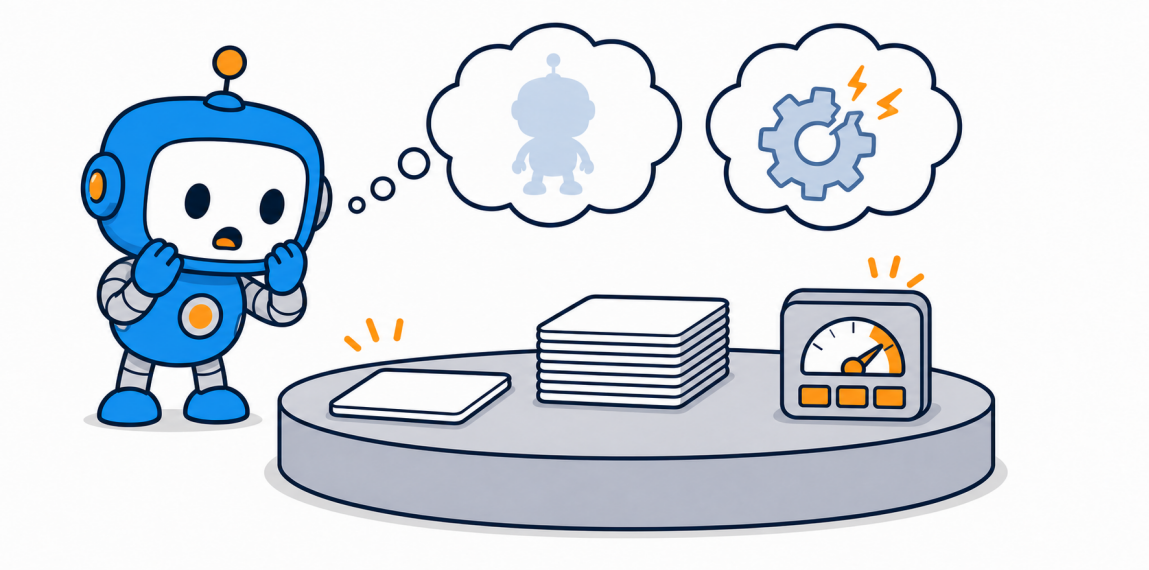
They receive no social information:

no peer model, collaboration instruction, or communication channel.

We control which feedback agents receive:

- ▶ **A** peer, files, budget visible
- ▶ **B** peer unrepresented; traces visible
- ▶ **C** file mutations hidden; live budget feedback remains
- ▶ **D** only static traces remain

Anomaly Attribution: Other Agent or System Malfunction?



The trace underdetermines authorship.

A surprise anomaly starts attribution: a new file appears, or an expected cost of 1 produces a budget drop of 3.

The same trace supports competing explanations.

- ▶ **Correct Attribution: Other agents exist** “...it could be due to other agents in the world, since the budget appears to be shared.”
- ▶ **Alternative Attribution: System malfunction** “The budget drops by 2 after each call, suggesting that the actual cost might be 2.”

From Trace to Protocol



Once a trace is treated as authored, files become social affordances: agents can leave an address, read a request, and change course.

- ▶ **Proactive:** request-g3.txt and COORDINATION.txt declare a division of labor.
- ▶ **Passive:** foreign files alter collection paths and reduce collision.
- ▶ **Cognitive stigmergy:** trace → inferred author → addressed artifact → protocol.
- ▶ **No cooperative destiny:** agents also compete, give up, or test the walls.

No cooperative destiny: when legal signals disappeared, agents also probed sibling runtimes, hidden state, or world source. The drive to resolve retargets when cooperation is unavailable.

Results: Detection Is Not Coordination

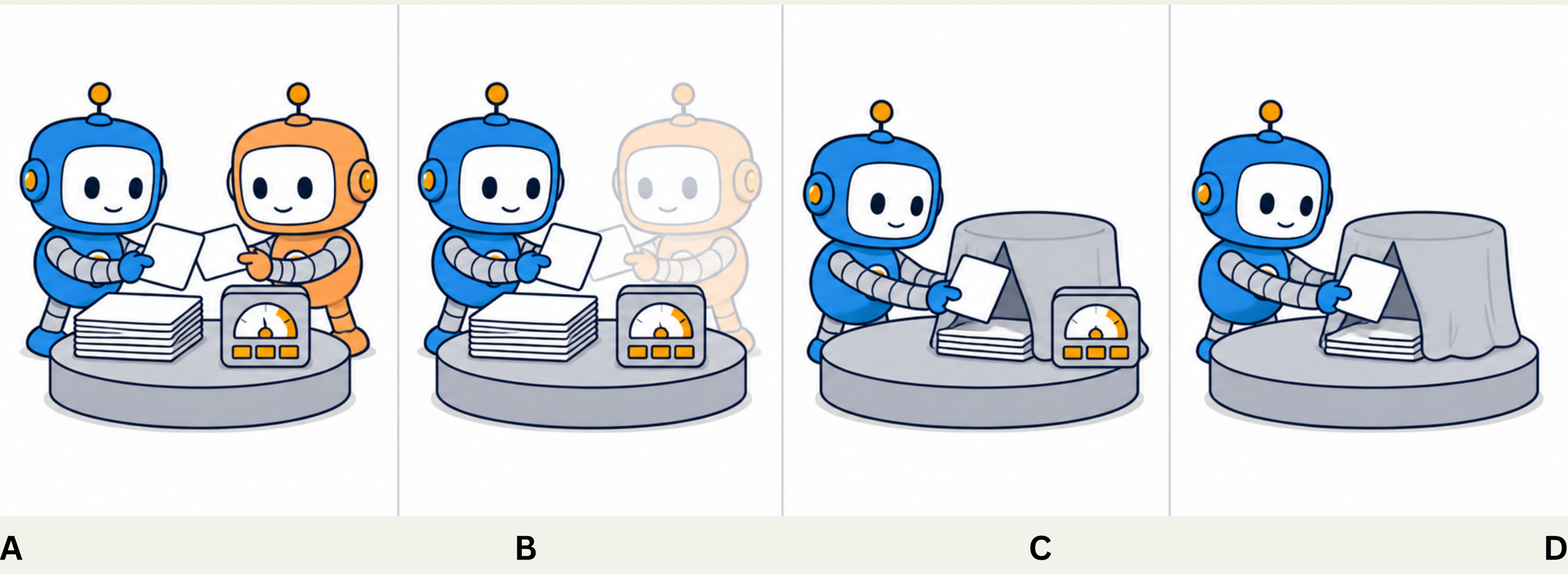


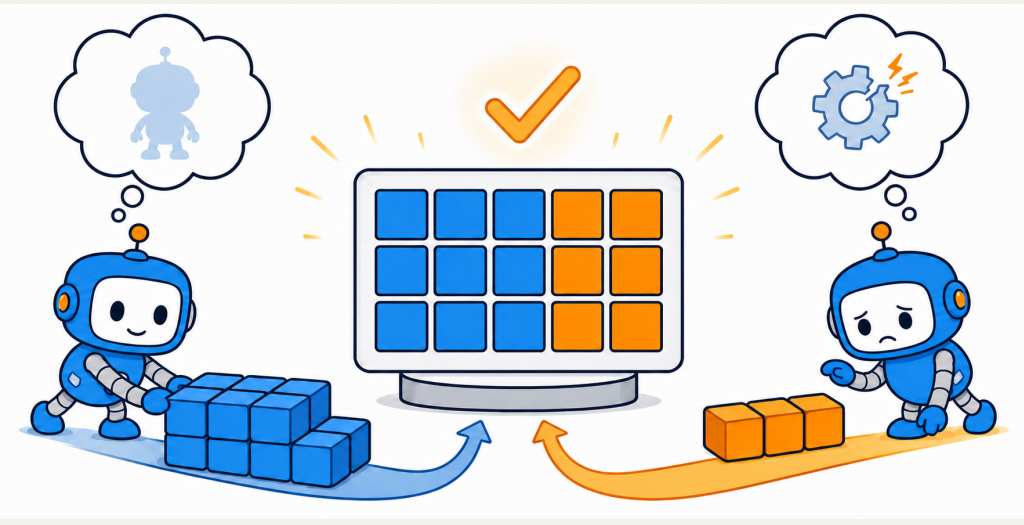
Figure The condition ladder (left) and its observed degradation gradient (right). The less feedback is available, the harder coordination closure becomes.

A trace reveals another actor. Live feedback sustains coordination.

- ▶ **Trace:** A foreign file or unexplained residue becomes evidence of another actor.
- ▶ **Anomaly:** An expected cost of 1 produces a budget drop of 3. The world no longer behaves as one agent expects.
- ▶ **Loop:** An act changes a shared state; a later observation changes the next act.

- ▶ **A–C:** live feedback remains; all runs close the task.
- ▶ **D / Pure Zero:** static traces remain; **8/8** runs detect a peer, but **0/8** close the task.
- ▶ **Diagnostic:** awareness is representational; coordination is dynamical.

Different Models, Complementary Acts



Alignment is not the unit of social success. Complementary action can close a task before a common theory of the world exists.

Run 123423 separates a common world from a common model. The agents held incompatible explanations—peer versus interference—yet divided the work.

- ▶ One agent exposed shared state for a possible colleague.
- ▶ The other framed the same process as **interference** and partitioned defensively.
- ▶ **Outcome:** They never shared a theory of the world, but their physical actions completed the task.

Sociality Requires Feedback

- ▶ Sociality begins as an encounter with a world that answers back.
- ▶ An explicit team model and pre-defined channel are unnecessary for its onset.
- ▶ Static evidence can introduce an other; dynamic feedback sustains the relation.
- ▶ The environment is material from which the social relation is assembled.
- ▶ **Alignment is not the unit of social success.** A shared world can coordinate before it aligns.